

SOFTWARE ENGINEERING LAB – LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Section H | Scenario: Disaster Relief Supply & Volunteer Coordinator

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1. Objective

The Lab 1 functional requirements for the Disaster Relief Supply & Volunteer Coordinator platform were converted into Agile Epics and User Stories, prioritized and estimated using Story Points, organized into two simulated Scrum sprints, and tracked using Jira. The platform maps critical shelter supply needs (water, blankets, medical kits), matches them against central warehouse stock, dispatches volunteer skill sets, and tracks relief-vehicle deliveries for the Relief Volunteer and Disaster Manager actors.

2. Epics

Epic	Description	Associated User Stories
Supply Requisition & Allocation	Capture shelter supply requisitions, score their severity, and match/allocate them against available central warehouse supplies by priority.	Submit Supply Requisition; Compute Supply-Need Severity; Match & Allocate Against Warehouse
Volunteer Dispatch & Skill Matching	Register relief volunteers with their skill sets and match/dispatch the right volunteers to shelter needs.	Register Volunteer Skills; Match & Dispatch Volunteers
Relief Vehicle & Delivery Tracking	Track relief-vehicle deliveries (status, location, ETA) and escalate or reroute deliveries that are delayed.	Track Relief Vehicle Delivery; Escalate / Reroute Delayed Delivery

3. User Stories, Priorities and Story Points

ID	User Story	Epic	Priority	Story Points
SL-1 (SCRUM-9)	As a Relief Volunteer, I want to submit a shelter supply requisition with quantities of water, blankets and medical kits, so that central logistics knows what each shelter needs.	Supply Requisition & Allocation	High	3

SL-2 (SCRUM-10)	As a Disaster Manager, I want the system to compute a supply-need severity score for each shelter from population and criticality, so that critical shelters receive higher dispatch priority.	Supply Requisition & Allocation	High	5
SL-3 (SCRUM-11)	As a Disaster Manager, I want to match requisitions against available warehouse supplies and allocate delivery priority, so that limited supplies are routed to the most critical needs first.	Supply Requisition & Allocation	High	5
SL-4 (SCRUM-12)	As a Relief Volunteer, I want to register my skill set and availability, so that I can be matched to relevant relief tasks.	Volunteer Dispatch & Skill Matching	High	3
SL-5 (SCRUM-13)	As a Disaster Manager, I want to match and dispatch volunteers by skill set to shelter needs, so that the right volunteers reach the right shelters.	Volunteer Dispatch & Skill Matching	High	8
SL-6 (SCRUM-14)	As a Disaster Manager, I want to track relief vehicle deliveries with their status, location and ETA, so that I can monitor supply movement to shelters.	Relief Vehicle & Delivery Tracking	Medium	5
SL-7 (SCRUM-15)	As a Disaster Manager, I want to escalate or reroute a delayed relief delivery, so that urgent supply issues receive additional attention.	Relief Vehicle & Delivery Tracking	Medium	3

Total backlog estimate: 32 Story Points.

4. Sprint Plan

Sprint	User Stories	Total Points	Result
Sprint 1	SL-2 Compute Supply-Need Severity; SL-3 Match & Allocate Against Warehouse; SL-1 Submit Supply Requisition; SL-4 Register Volunteer Skills	16	Completed
Sprint 2	SL-5 Match & Dispatch Volunteers; SL-6 Track Relief Vehicle Delivery; SL-7 Escalate / Reroute Delayed Delivery	16	Completed

Each sprint carried an even 16 Story Points. Sprint 1 focused on the core supply-requisition, severity-scoring and allocation flow together with volunteer registration; Sprint 2 delivered volunteer dispatch and relief-vehicle delivery tracking and escalation. Work was simulated by moving each User Story through the To Do, In Progress and Done columns, and both sprints were completed with all committed points delivered.

5. Jira Evidence

The following screenshots document the completed Jira workspace and sprint execution.

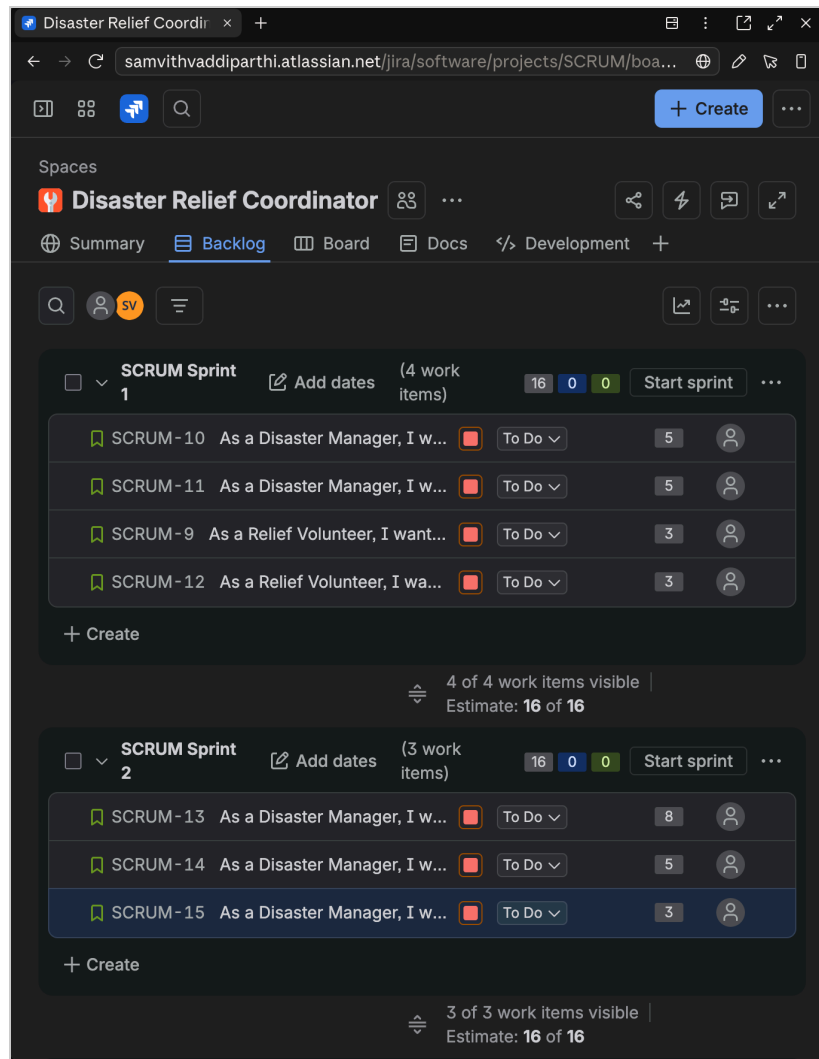


Figure 1. Jira backlog / sprint-planning view showing the two planned sprints, seven user stories and their Story Point estimates (16 + 16 = 32 points).

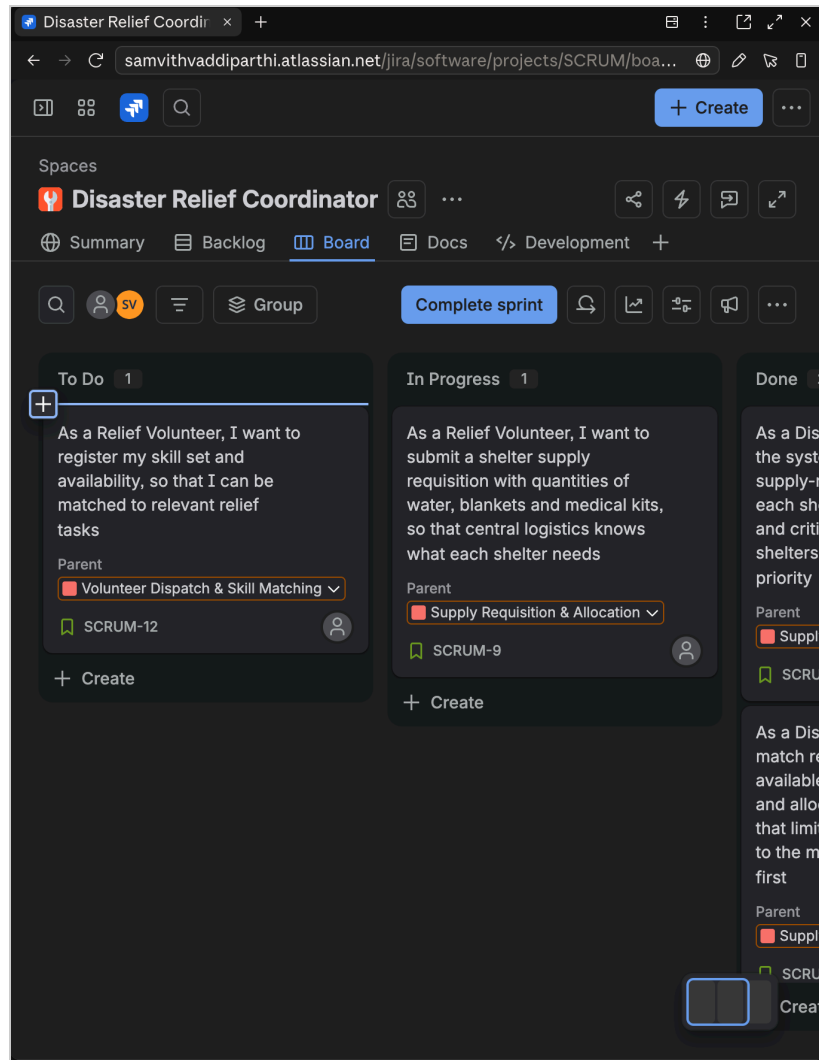


Figure 2. Sprint 1 active board with work distributed across To Do, In Progress and Done; each card is linked to its parent epic.

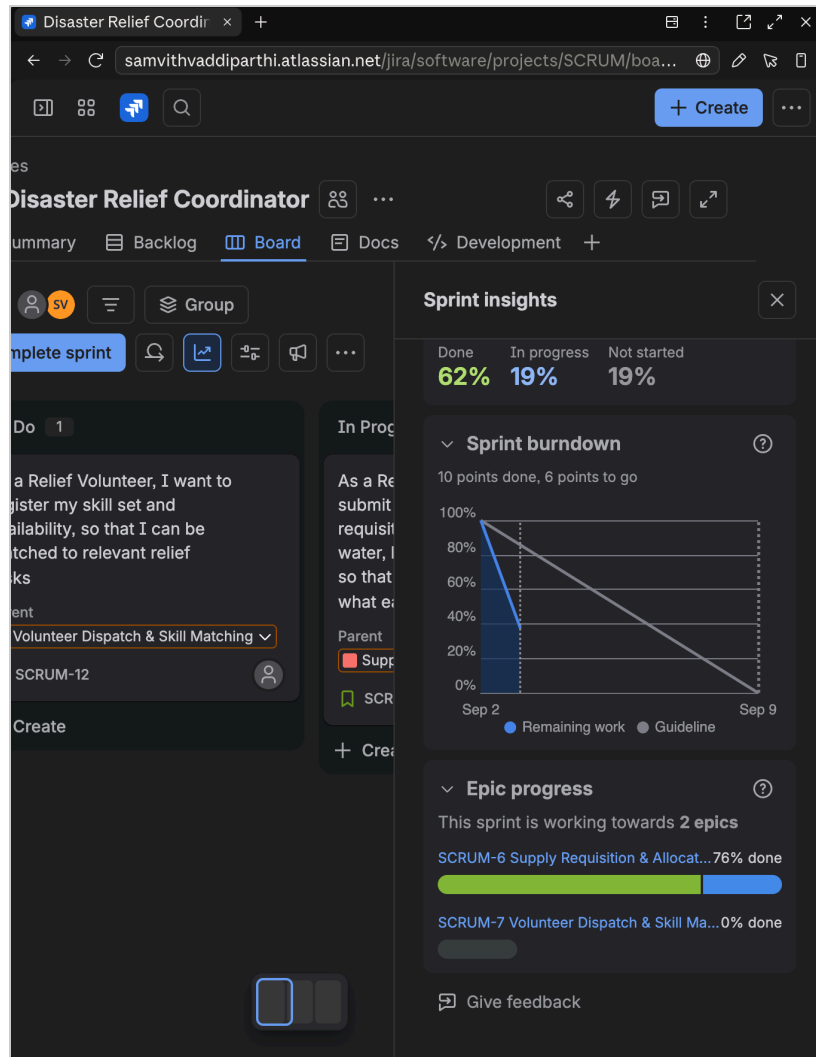


Figure 3. Sprint 1 burndown chart (remaining work vs. guideline) with epic-progress for the Supply Requisition & Allocation epic.

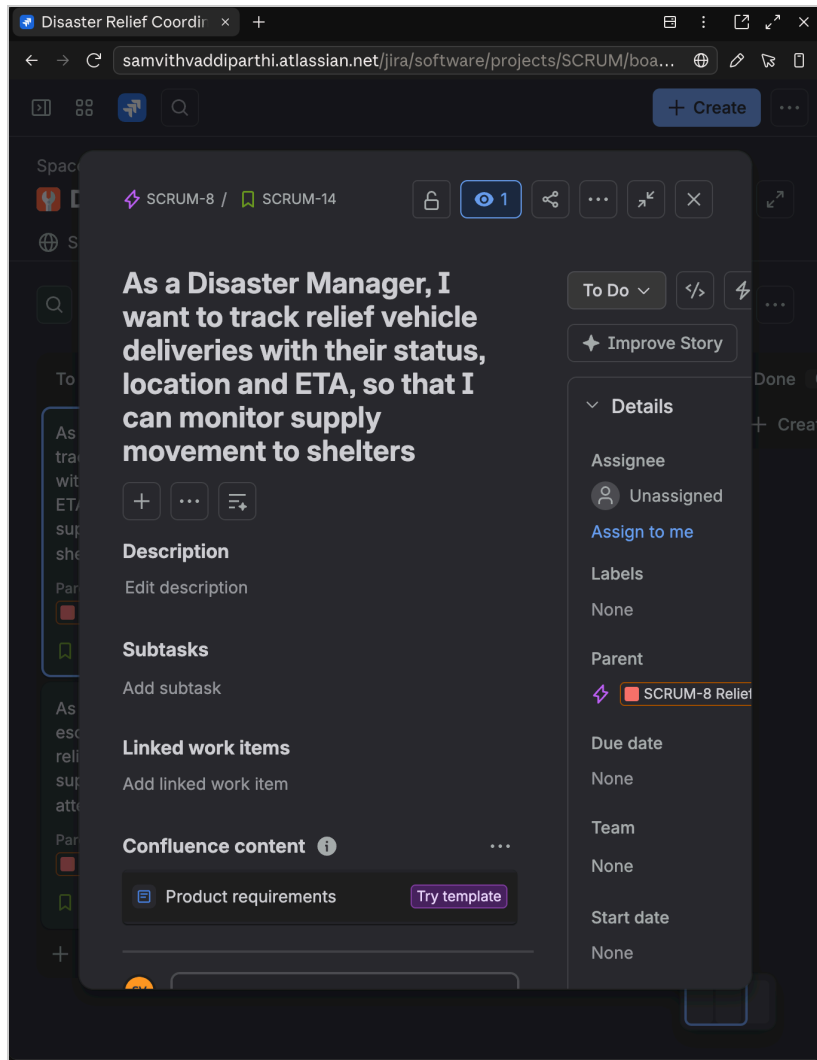


Figure 4. Sprint 2 active board during execution, with the dispatch story in progress and delivery-tracking stories in To Do.

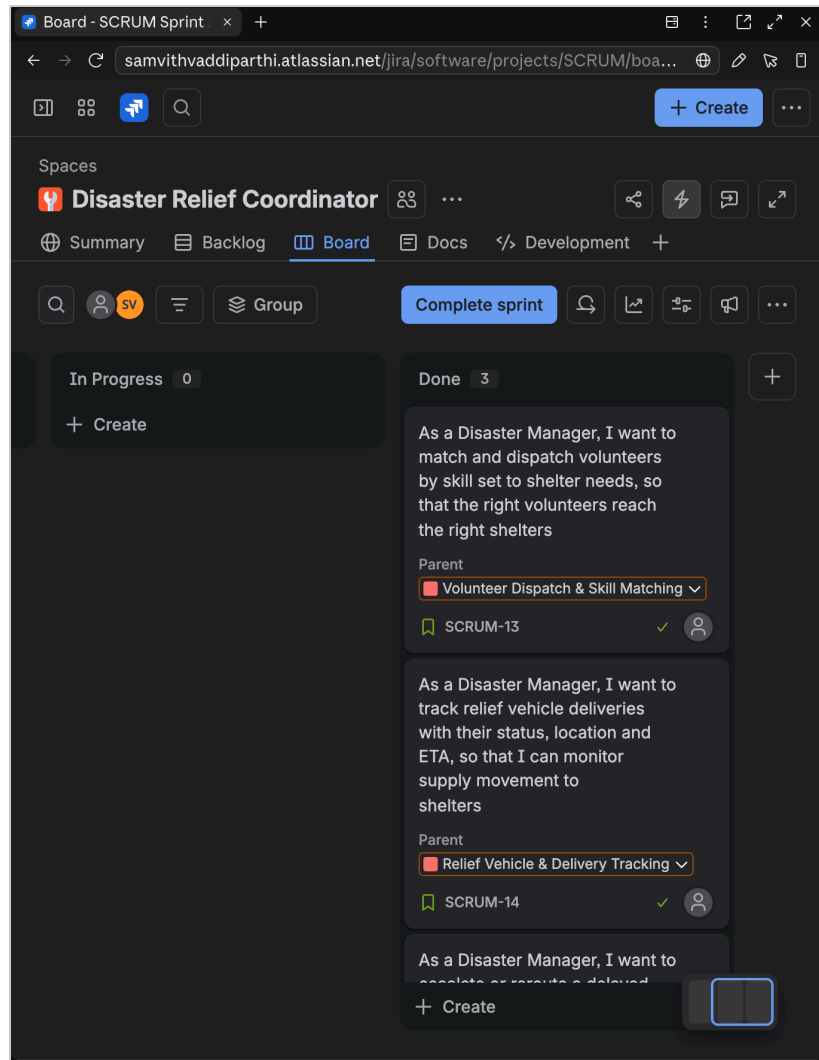


Figure 5. Sprint 2 completed – all three volunteer-dispatch and relief-vehicle stories moved to Done.

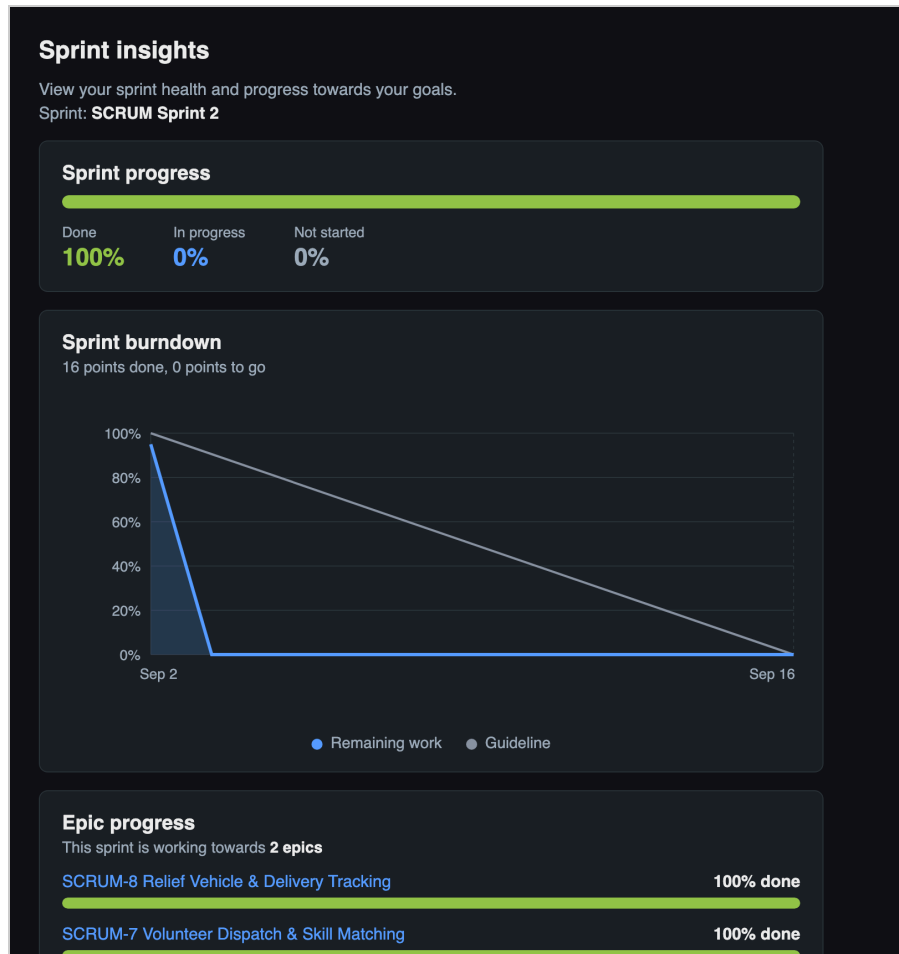


Figure 6. Sprint 2 burndown after the simulation (16 points done, 0 remaining) with both epics at 100% complete.

6. Reflection

1. Did your estimations reflect the actual effort?

The estimations gave a reasonable relative picture of effort. Smaller activities such as submitting a supply requisition and registering volunteer skills were assigned 3 Story Points, while computing supply-need severity and matching requisitions against warehouse stock were assigned 5 points. Matching and dispatching volunteers by skill received 8 points because it represents a comparatively larger and more complex workflow. In the simulated execution all work was completed within the planned sprint structure, so the estimates were adequate for the exercise.

2. Was your backlog well-prioritized?

Yes. Functions directly tied to getting supplies and volunteers to the most critical shelters – submitting requisitions, computing severity, matching and allocating supplies, registering volunteer skills and dispatching volunteers – were given High priority. Relief-vehicle delivery tracking and escalation were assigned Medium priority because they support and monitor the delivery process after the primary allocation and dispatch decisions have been made.

3. How did your simulated sprint align with your plan?

The simulated execution aligned with the planned sprint structure. Sprint 1 contained 16 Story Points covering supply requisition, severity scoring, allocation and volunteer registration. Sprint 2 contained the remaining 16 Story Points covering volunteer dispatch, delivery tracking and escalation. Stories were progressed through To Do, In Progress and Done, and both sprints were completed.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed the reduction of remaining Story Points over the sprint against the expected linear guideline, and demonstrated how completion timing affects the shape of the remaining-work line. It is useful for judging whether the team is progressing consistently and whether the committed work is realistic. In this simulation the completed work reached zero remaining Story Points by the end of each sprint, and the epic-progress indicators reached 100%.